Weighted orbit modes and direct phase extraction

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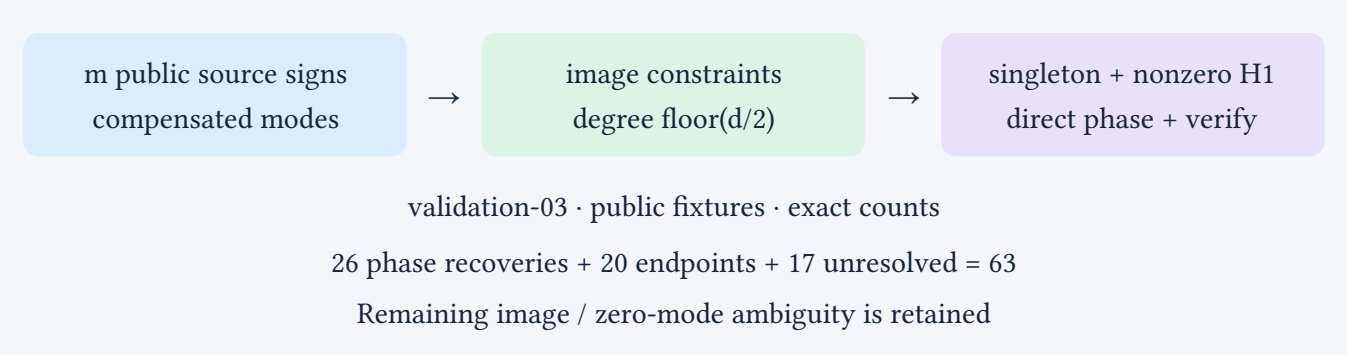
The compensated modes satisfy $F_e = Ax^e H_{e(v)}$, where $A = m\chi(b)$ and $v = x^m$. Roots-of-unity filtering proves the identity for both parities of $d = \frac{r-1}{m}$. For $d \geq 2$, the zeroth mode bounds the image candidate count by $\lfloor \frac{d}{2} \rfloor$.

If the image is known and $H_1(v) \neq 0$, recover $x = \frac{F_1}{AH_1(v)}$, then verify the public-point equation. A zero denominator remains unresolved. The second mode supplies a division-free phase-cancellation constraint.



Exact finite checks. 44 small configurations: 28 even-d and 16 odd-d; 15,321 mode identities; 4,704 conditional phase extractions; 822 zero-denominator cases retained. Four larger coefficient compilers checked at r=65,537. Independent small dense powers and streamed larger references are distinguished in DERIVATION.md.

Public recovery. Four prescribed radii, no scalar labels at runtime. Of 63 fixtures, 46 recover: 26 phase extractions and 20 endpoints. All 17 unresolved cases belong to the frozen F509/r479/m2 panel. Source-call count is 527, including target validation; source setup and final verification are separately required. These are exact counts, not timing measurements.

Fixture	Cases	Recovered	Unresolved
F7/r13/m3	13	13	0
F7/r13/m6	13	13	0
F509/r479/m2	20	3	17
F701/r683/m11	17	17	0

Costs and scope. Scalar product/remainder trees compile selected coefficients; conservative field work is O(M(m+d) log(m+d)+log r). Each orbit still queries m source signs. Source domain restrictions and the original 4,563-case cohort are unchanged. No full formal proof or signed independent review is supplied.

Next: compose mode constraints and direct phase extraction with retained-sign fiber recovery; investigate the remaining quadratic-image ambiguity. No automatic fallback was added to hide the unresolved cases.

Evidence: modes.rs, receipt.json, check.stdout, validation-01 and validation-02 snapshots. Sources: Bostan–Gaudry–Schost (2007), specfun.inria.fr/bostan/publications/BoGaSc07.pdf; FLINT gr_dft transform conventions. KB unavailable; bounded local and primary-source fallback recorded.